TWO IDEALS WITH TRIVIAL SUM CAN BE INTERPOLATED BY A UNIT-VALUED GROUP ELEMENT WITH PRESCRIBED RESIDUES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the element x and the change of ideals, the reformulation as a unit condition, and the application of the monogenic-extension lemma.

Editorial note. The label lemma-interpolate names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Absolutely integrally closed A-algebras and the group G(R). Same chapter, the definition of an absolutely integrally closed ring and of the group G(R) of unit-valued elements used in the statement. Supplies the ambient ring, the ideals, the elements sigma_0, sigma_1 and the group in which g is sought.
- The extension lemma for a pair of coprime ideals. Same chapter, the lemma on finding a monogenic extension with a unit of the form a - by, applied here to the reduction problem. Supplies the existence of the interpolating element once the problem is reduced to a unit condition.

Source reference. Original reference 0GY2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14523–14523. Prerequisite pointer invoked at proof line(s) 14726; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be an A-algebra which is absolutely integrally closed. Let $I, J \subset R$ be ideals with $I + J = R$. There exists a $g \in G(R)$ such that $g \bmod I = \sigma_0$ and $g \bmod J = \sigma_1$.*

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Proof. Choose $x \in I$ such that $x \equiv 1 \bmod J$. We may and do replace I by xR and J by $(x-1)R$. Then we are looking for an $s \in R$ such that

(1) $1 + \pi s$ is a unit,
(2) $s \equiv 0 \bmod xR$, and
(3) $s \equiv 1 \bmod (x-1)R$.

The last two conditions say that $s = x + x(x-1)y$ for some $y \in R$. The first condition says that $1 + \pi s = 1 + \pi x + \pi x(x-1)y$ needs to be a unit of R. However, note that $1 + \pi x$ and $\pi x(x-1)$ generate the unit ideal of R and that $1 + \pi x$ is an ℓth root of 1 modulo $\pi x(x-1)$[1]. Thus we win by Lemma 0GY2 and the fact that R is absolutely integrally closed. □

[1]Because $1 + \pi x$ is congruent to 1 modulo π, congruent to 1 modulo x, and congruent to $1 + \pi = \zeta$ modulo $x - 1$ and because we have $(\pi) \cap (x) \cap (x-1) = (\pi x(x-1))$ in $A[x]$.